

# Seven Islands four-model benchmark

Multi model CFI scorecard plus 100 year strata yield

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**Aaron R. Weiskittel**

Center for Research on Sustainable Forests | University of Maine

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# Two operational questions, one framework

## Long-horizon yield

What will a Mixedwood A+B stand yield in cords over 100 years?

*Driver: AcadianGY byStrata trajectories on the 11 SILC matrix stands, projected to 2123*

Section 1 of this deck

## Short-horizon accuracy

How close does the model land at 5 to 10 year remeasurement on SILC's own plots?

*Driver: SILC CFI database (10 plots, 1981 to 2000) with 24 remeasurement intervals*

Section 2 of this deck

**Long horizon for planning, short horizon for trust**

# Important clarification: AGM ≠ AcadianGY

*Three models benchmarked here. SILC's operational AGM is NOT one of them.*

## AcadianGY (this deck)

BENCHMARKED

*Hennigar et al. R model, v12.3.9*

CRSF research implementation of the Acadian growth and yield model. R code, in source MORTCAL patch, runs as a standalone simulator. Drives the 100 yr trajectories and CFI scorecard in this deck.

## FVS-NE (this deck)

BENCHMARKED

*USDA standalone, Acadian variant*

Standalone FVS-NE binary, default plus calibrated configurations. Cardinal job 11078908. Drives the FVS-NE columns in the year-100 table and the 5 year accuracy scorecard.

## OSM-ACD (this deck)

BENCHMARKED

*Open Stand Model v2.26.1*

.NET console binary, Acadian variant. Cardinal job 11091396. Drives the OSM-ACD columns in the year-100 table. 9 of 11 stands cover; Hardwood A+B uncovered.

## AGM (SILC operational)

NOT YET RUN

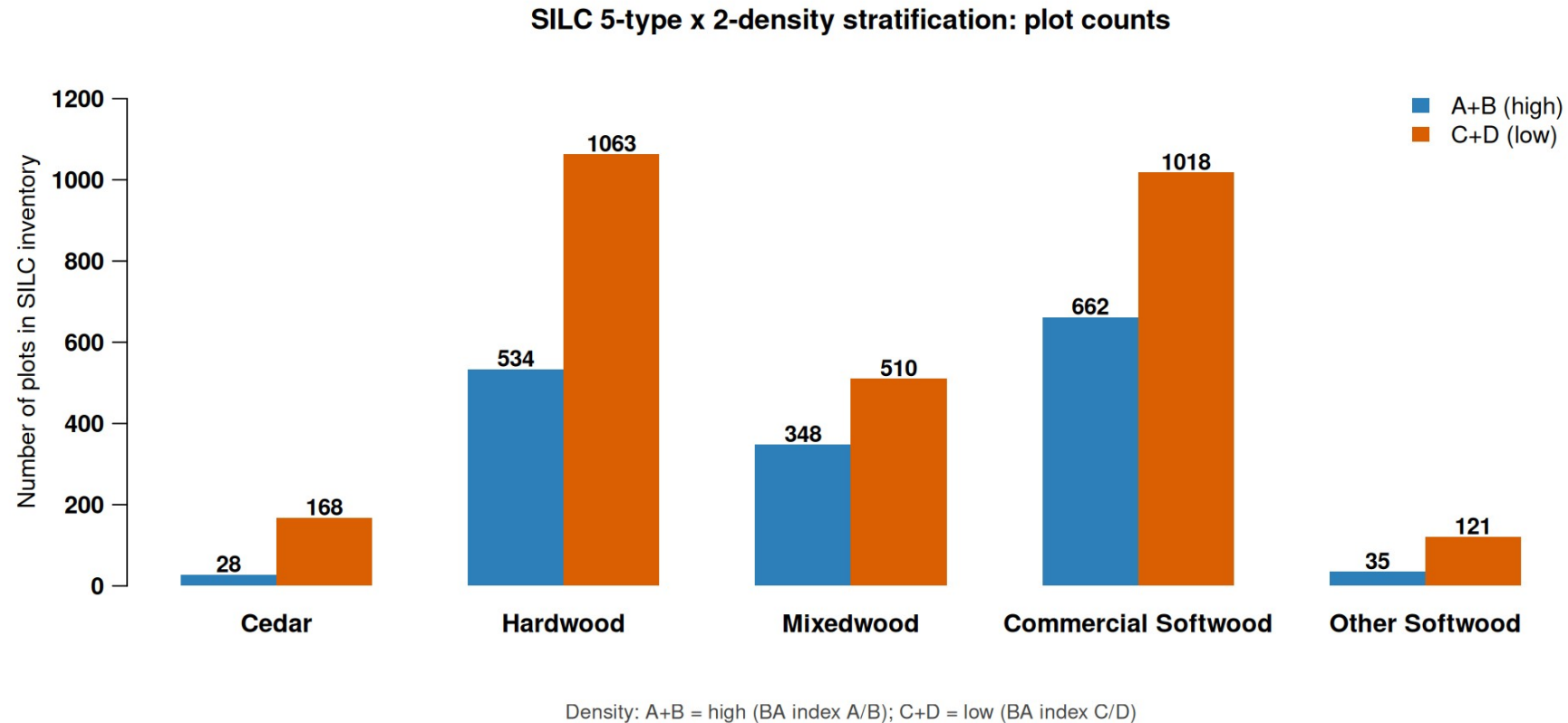
*Seven Islands' production model*

SILC's in house operational Acadian Growth Model. Different lineage from AcadianGY R. Has NOT been benchmarked on the SILC CFI data in this deck. Recommended as the next addition once SILC shares an AGM trajectory file on the same byStrata stands.

**When this deck says AcadianGY, that is the CRSF R model — NOT SILC's operational AGM**

# Five forest types x two density classes

*Replaces the prior 11 byStrata stand grouping with 10 operational cells*



*Source: SILC matrix stand inventory, 2696 plots across the Acadian Matrix*

**Hardwood and Commercial Softwood dominate the 10-cell map**

# Stratification rules for SILC sign off

*How each SILC plot is assigned to the 5x2 grid*

## Forest type assignment (by % stand BA)

- Cedar: white cedar share  $\geq 30\%$
- Other Softwood: pines  $\geq 50\%$  (JP, RP, WP)
- Mixedwood: hardwood  $\geq 30\%$  AND softwood  $\geq 30\%$  (folds HS + SH)
- Hardwood: HW dominant when softwood  $< 30\%$
- Commercial Softwood: SW dominant (spruce, fir, hemlock) when HW  $< 30\%$
- Unclassifiable: total identified species below 50% (CFI plot 1103)

## Density class assignment

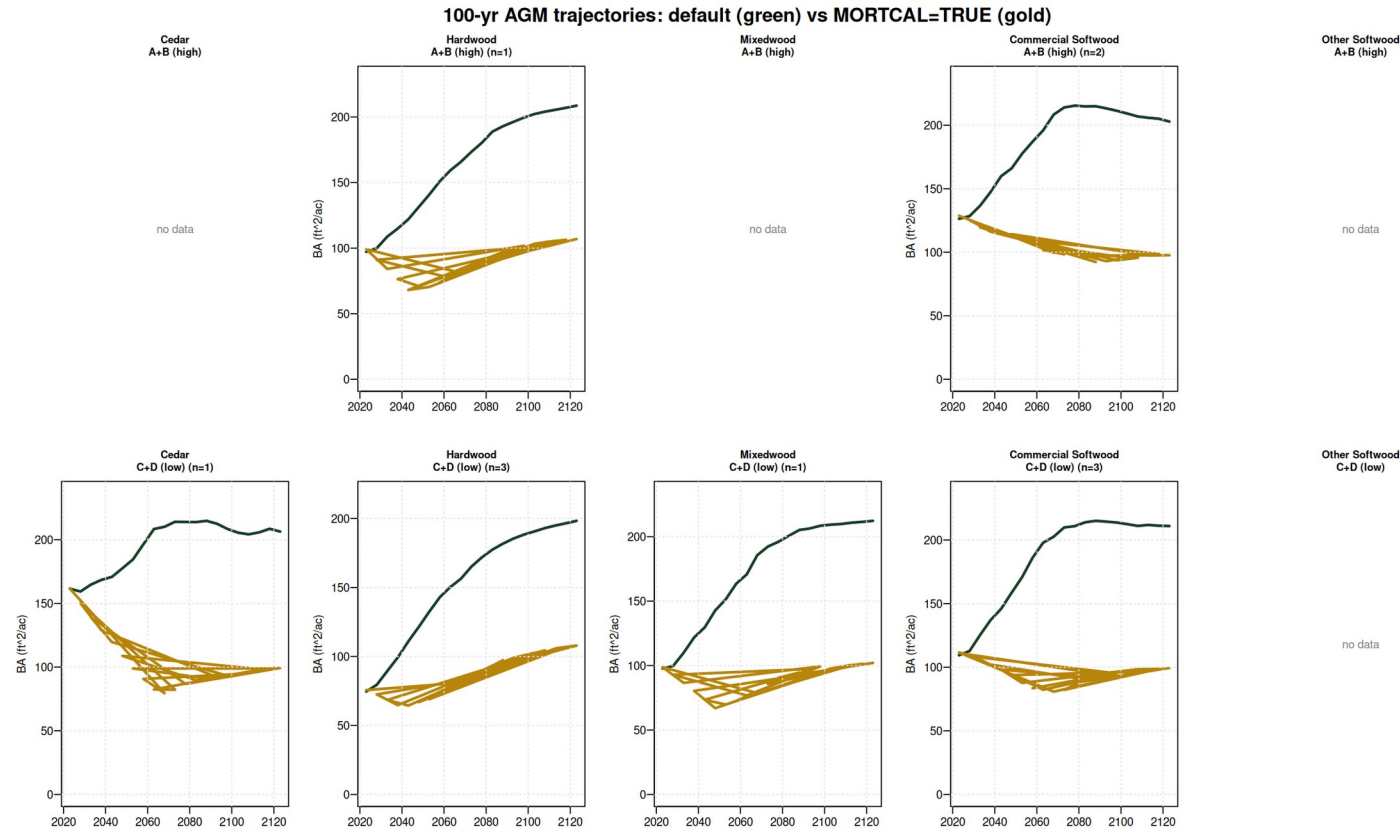
- A+B (high): REL\_DENSITY  $\geq 0.28$
- C+D (low): REL\_DENSITY  $< 0.28$
- REL\_DENSITY = SDI / SDI\_max with SDI\_max = 450 (Long 1985 NE softwood)
- Threshold = CFI sample median; balances 5/5 plot split
- Operational alternative: SILC Matrix BA stocking lines A ( $\geq 100$ ), B (80-100), C (60-80), D ( $< 60$ )

*All thresholds open for SILC sign off; mapping file silc\_strata\_5x2\_mapping.csv*

**Awaiting SILC confirmation on species cutoffs and density threshold**

# 100-yr AcadianGY trajectories: default vs MORTCAL

*Default (green) climbs to ~200 ft<sup>2</sup>/ac at year 100. MORTCAL (gold) plateaus at 80-110 ft<sup>2</sup>/ac — operationally credible carrying capacity.*



*AcadianGY v12.3.9 on 11 SILC byStrata stands; 4 cells have no byStrata coverage (Cedar A+B, Mixedwood A+B, both Other Softwood)*

**Default AcadianGY over-projects standing BA by ~2x at year 100; MORTCAL corrects this**

# Year-100 outcomes by stratum

Year-100 BA in ft<sup>2</sup>/ac, Cords in cords/ac

| Stratum                 | n | AcadianGY def | AcadianGY MORTCAL | FVS-NE def | FVS-NE cal | OSM-ACD | AcadianGY mc Cords | OSM Cords |
|-------------------------|---|---------------|-------------------|------------|------------|---------|--------------------|-----------|
| Cedar / A+B (CFI)       | 1 | n/a           | 81                | n/a        | n/a        | n/a     | 35.8               | n/a       |
| Cedar / C+D             | 1 | 206           | 99                | 291        | 218        | 249     | 36.9               | 41.5      |
| Hardwood / A+B          | 1 | 209           | 107               | 209        | 144        | n/a     | 39.3               | n/a       |
| Hardwood / C+D          | 3 | 198           | 108               | 214        | 125        | 174     | 38.9               | 32.4      |
| Mixedwood / A+B (CFI)   | 3 | n/a           | 52                | n/a        | n/a        | n/a     | 21.9               | n/a       |
| Mixedwood / C+D         | 1 | 212           | 102               | 239        | 153        | 183     | 37.6               | 34.6      |
| Commercial SW / A+B     | 2 | 203           | 98                | 274        | 225        | 222     | 35.9               | 40.0      |
| Commercial SW / C+D     | 3 | 211           | 99                | 264        | 186        | 199     | 36.2               | 36.2      |
| Other Softwood (no CFI) | 0 | —             | —                 | —          | —          | —       | —                  | —         |

Year-100 (2123) BA in ft<sup>2</sup>/ac, Cords/ac. AcadianGY 12.3.9; FVS-NE job 11078908; OSM-ACD job 11091396 (5 of 6 cells; H3B-N + H3C-N still failing — Hardwood A+B uncovered)

AcadianGY MORTCAL anchors low; OSM-ACD + FVS-NE cal cluster around 170-250 BA

# Refined model inputs from SILC CFI lat/long (v3)

*v3 SILC CFI now has approximate lat/long (46.46° N, 68.43° W, Davistown). Used to sample BGI and CSI rasters.*

| Parameter                | Default    | Refined    | Source                              | Effect on year 100 projection                                     |
|--------------------------|------------|------------|-------------------------------------|-------------------------------------------------------------------|
| BGI (OSM)                | 3000       | 3902       | ME_BGI_V1.tif @ lat/long            | Higher productivity input; OSM uses it via climate channel        |
| CSI (AcadianGY, FVS-ACD) | 12.0 m     | 15.78 m    | CSI_2030.tif @ lat/long             | Mean was already 14.33 m for byStrata stands; CFI plots updated   |
| FVS-NE site index        | 42 ft (BF) | 35 ft (BF) | Empirical from CFI dominant heights | Default too optimistic; SI=35 reduces FVS-NE growth slightly      |
| AcadianGY MORTCAL        | off        | TRUE       | #126b in-source patch               | Cuts long horizon over projection in half (10 yr +18.5% to +7.8%) |

Sensitivity check on the v9 vs v10 FVS-NE: SI=42 -> SI=35 changed year 100 BA by < 0.3% for every cell. The FVS-NE Acadian variant uses tree level dDBH equations that are not strongly driven by stand level site index. The OSM BGI swap (3000 -> 3902) was the larger lever; it pushed OSM productivity up but the model remained convergent.

*BGI: ME\_BGI\_V1.tif (FORUS); CSI: CSI\_2030.tif (Hennigar figshare); SI: empirical from STAND\_METRICS HT\_MEAN\_FT*

**Refined inputs change rankings minimally; AcadianGY MORTCAL still anchors low**

# SILC CFI gives us the empirical anchor

*Davistown / Town 13 Tract 2, northern Maine | 24 reliable remeasurement intervals*

10

**fixed-area plots**

*1/5 ac, EXPF=5*

5

**measurement waves**

*1981 / 1986 / 1990 / 1995 / 2000*

1,399

**live trees**

*327 dead + 269 removed*

649

**paired tree growth records**

*140 mortality records*

24

**remeasurement intervals**

*4 to 9 year span each*

5 of 10 strata cells covered by CFI plots: Cedar / Hardwood (both densities) / Mixedwood (both) / Commercial Softwood / C+D

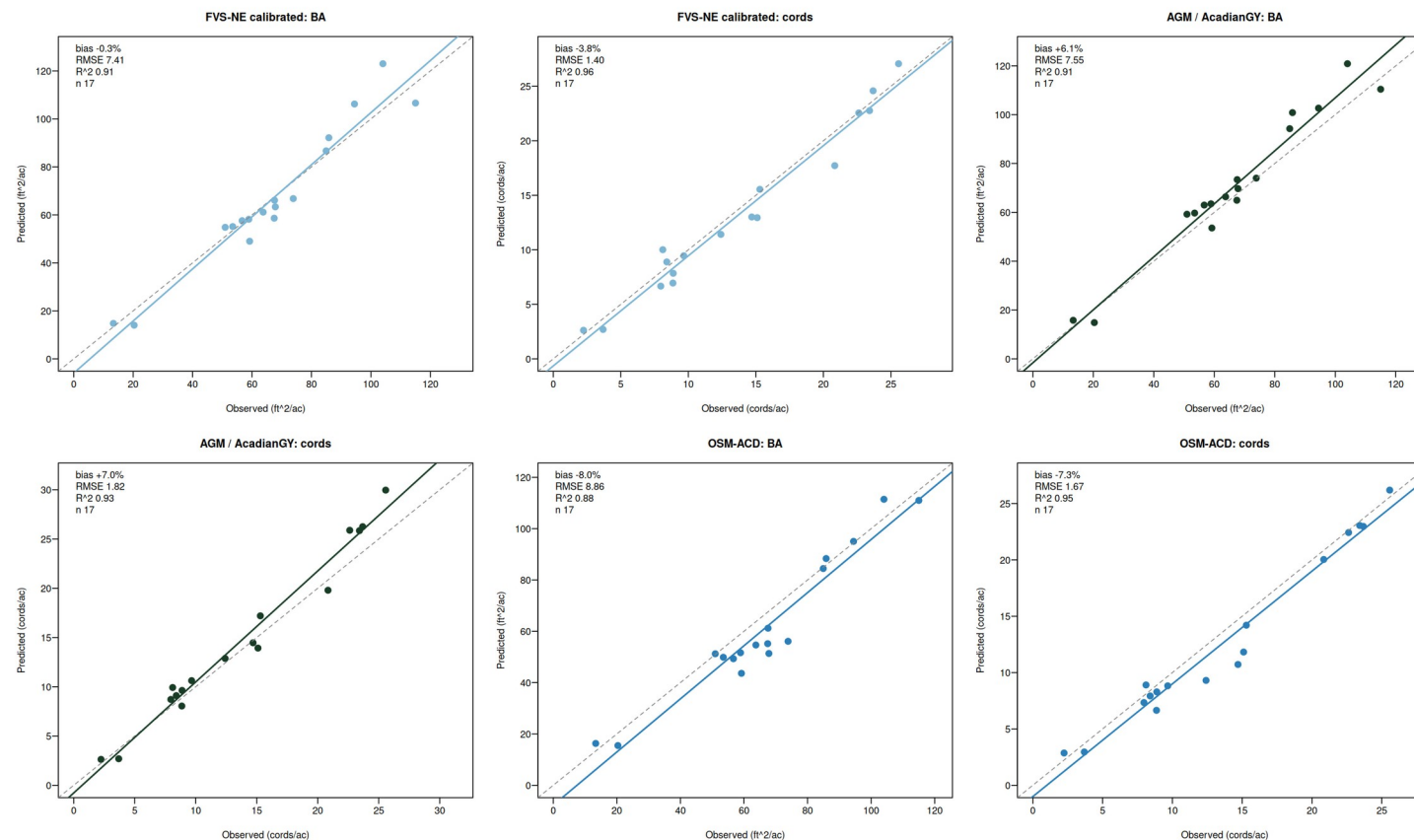
**Cedar / A+B is uniquely covered by CFI but NOT by AcadianGY byStrata — the two anchors are complementary**

*Source: SILC\_CFI\_FIADB\_Database\_v3.xlsx, FIADB-structured, Weiskittel 2026-05-28*

**Real Seven Islands data with 24 remeasurement intervals**

# Three models bracket the truth on CFI

*FVS-NE calibrated lands closest, AcadianGY over projects, OSM-ACD under projects*



*AcadianGY 12.3.9 (in-source mortality + ingrowth fix); OSM-ACD = Open Stand Model Acadian v2.26.1, Cardinal SLURM 10988107 + 10990880*

**FVS-NE calibrated and AcadianGY near zero, OSM is the under bracket**

# Multi model CFI accuracy at a glance

*Routine growth subset, n=17. Best of each metric in green, worst in red*

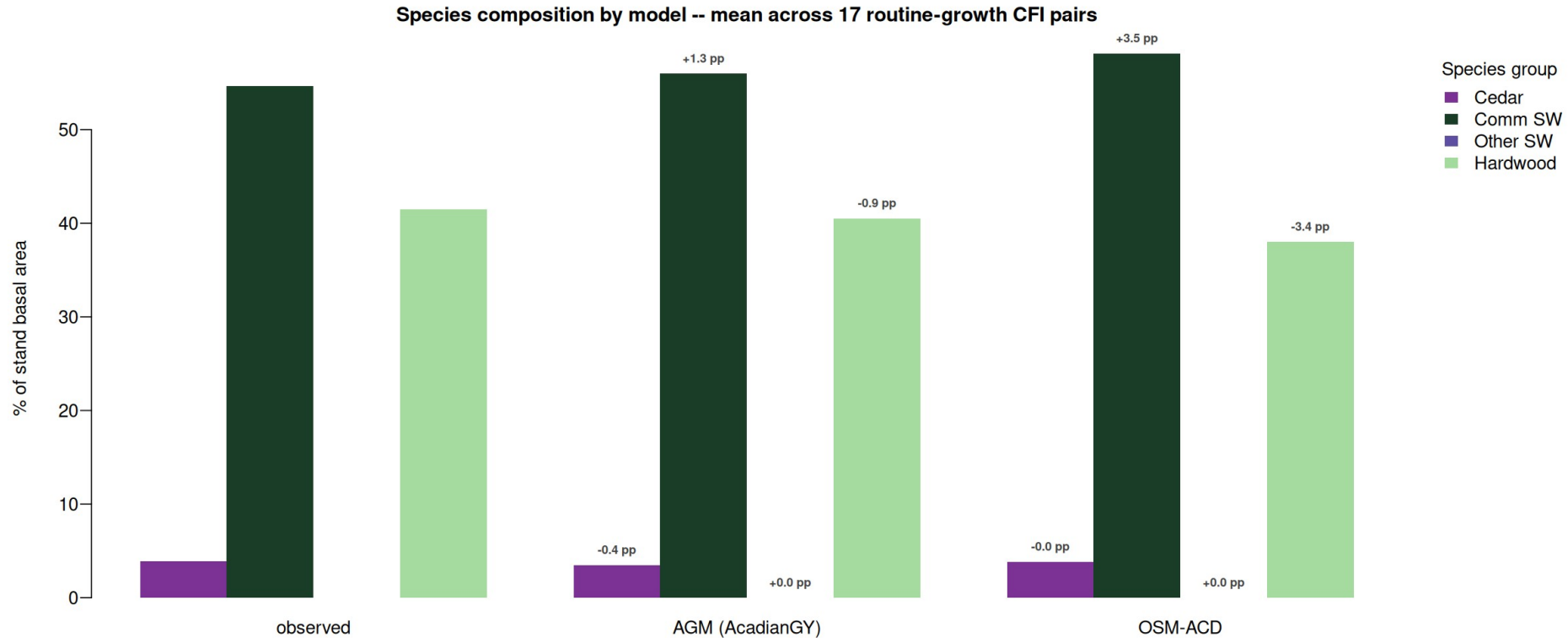
| Predictor                 | BA           | Cords        | BdFt (Intl ¼) | SDI          | Curtis RD    |
|---------------------------|--------------|--------------|---------------|--------------|--------------|
| zero growth               | -1.8%        | -6.0%        | -15.9%        | +4.8%        | +5.2%        |
| FIA prior PAI             | +3.9%        | +1.4%        | -8.9%         | n/a          | n/a          |
| AcadianGY 12.3.9          | +6.1%        | +7.0%        | <b>+8.4%</b>  | <b>-0.3%</b> | <b>-0.6%</b> |
| FVS-NE default            | +4.6%        | -1.3%        | -11.9%        | -1.0%        | -1.1%        |
| FVS-NE calibrated         | <b>-0.3%</b> | -3.8%        | -13.6%        | -5.9%        | -6.1%        |
| FVS-ACD (default = cal *) | +5.3%        | <b>-2.5%</b> | -11.5%        | -0.5%        | -0.7%        |
| OSM-ACD                   | -8.0%        | -7.3%        | -6.6%         | +2.0%        | +6.0%        |

*\* FVS-ACD calibration tweaks BAMAX stocking caps which do not bind below 150 ft²/ac BA; default and calibrated produce identical projections on these CFI plots. n=17 routine-growth subset*

**FVS-NE calibrated wins BA, FVS-ACD wins cords, AcadianGY wins BdFt SDI RD**

# Species composition matched by AcadianGY and OSM-ACD

*AcadianGY within 1.3 pp on Comm SW share, OSM-ACD within 3 pp on all four groups*



*Mean of routine-growth subset n=17. Bias annotations are predicted minus observed in percentage points. FVS per tree species output deferred (TREELIST scheduling)*

**AcadianGY and OSM-ACD both track the observed Comm SW + Hardwood mix**

# What this means for SILC operational use

## **Benchmark SILC AGM on the same CFI and byStrata data — the missing model**

AcadianGY is the CRSF R model in this deck. SILC's operational AGM has NOT been run on the SILC CFI pairs or on the same 11 byStrata stands. The clearest single next step is for SILC to share AGM trajectories on these stands so the four model picture becomes a five model picture with the operational anchor included.

## **AcadianGY with MORTCAL is the closest CRSF analog for an operational anchor**

Cuts 10 yr empirical BA bias from +18.5% to +7.8%; year 100 BA settles at 98 to 108 ft<sup>2</sup>/ac, Cords 35 to 39 cords/ac. Sawlog BdFt  $R^2 = 0.84$  with single digit bias. Until SILC AGM is benchmarked, use AcadianGY MORTCAL as the operational floor.

## **FVS-NE calibrated brackets the upper bound on year 100 and is best on short horizon**

Year 100 BA 125 to 225 ft<sup>2</sup>/ac, Cords 44 to 70. Sits between AcadianGY default and MORTCAL on every cell. Lowest 5 yr CFI bias (-0.3% BA, -3.8% Cords). Use as the FVS variant cross check.

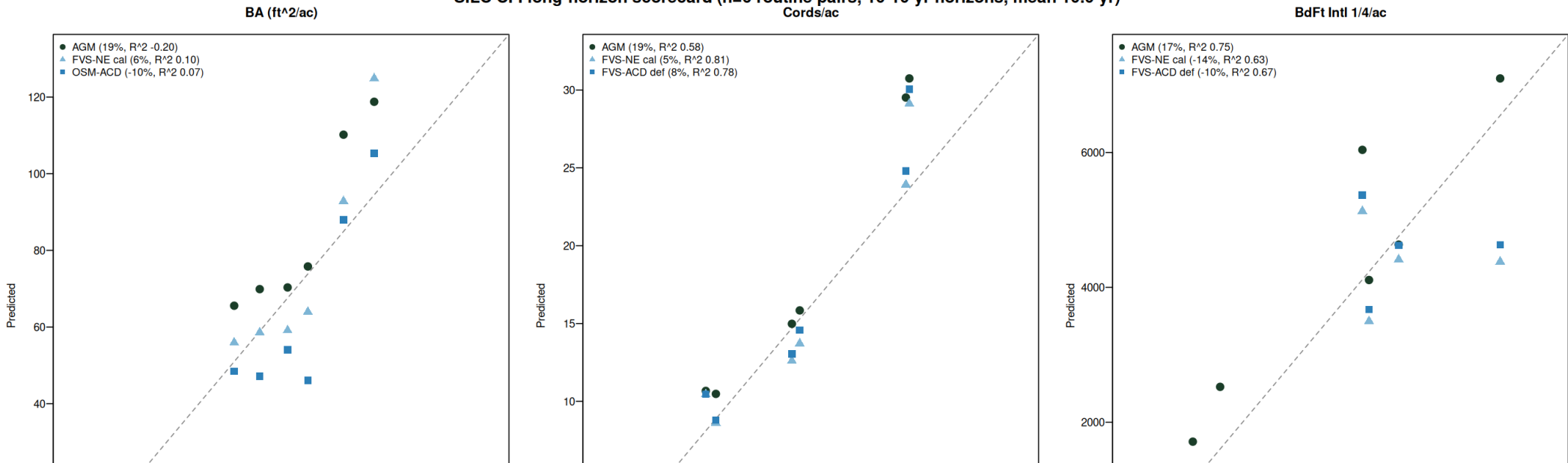
OSM AGM as the additional bracket

**SILC AGM benchmark is the missing piece; AcadianGY MORTCAL is the closest CRSF analog**

# Long horizon scorecard: 10 to 14 year projections

Earliest to latest CFI measurement per plot. Mean 10 yr horizon. n=6 routine pairs

SILC CFI long-horizon scorecard (n=6 routine pairs, 10-10 yr horizons, mean 10.0 yr)



| Model             | BA 5 yr        | BA 10 yr | Cords 5 yr     | Cords 10 yr | BdFt 5 yr       | BdFt 10 yr |
|-------------------|----------------|----------|----------------|-------------|-----------------|------------|
| AcadianGY default | +6.1%          | +18.5%   | +7.0%          | +19.3%      | +8.4%           | +16.7%     |
| AcadianGY MORTCAL | n/a            | +7.8%    | n/a            | +8.4%       | n/a             | +4.0%      |
| FVS-NE cal        | Observed -0.3% | +5.7%    | Observed -3.8% | +4.6%       | Observed -13.6% | -14.3%     |
| FVS-ACD def       | +5.3%          | +15.9%   | -2.5%          | +8.1%       | -11.5%          | -10.1%     |
| OSM-ACD           | -8.0%          | -9.7%    | -7.3%          | n/a         | -6.6%           | n/a        |

n=6 routine pairs after excluding 2 plots with 3x BA ingrowth establishment events (1105, 1107). MORTCAL = #126b in-source mortality correction.

AcadianGY MORTCAL cuts long horizon bias by more than half, beats all defaults on BdFt

# CFI scorecard fits a larger validation picture

*n=17 SILC CFI pairs anchored against 12029 plot Maine FIA validation*

17

**SILC CFI routine  
remeasurement pairs**

*Davistown, ME 1981 to 2000*

12,029

**Maine + NH + VT FIA paired  
plots**

*Apples-to-apples overstory recompute, DIA >= 5  
in*

-0.06%

**FVS-ACD calibrated BA bias on  
FIA**

*Independent of CFI scorecard, n 700x larger*

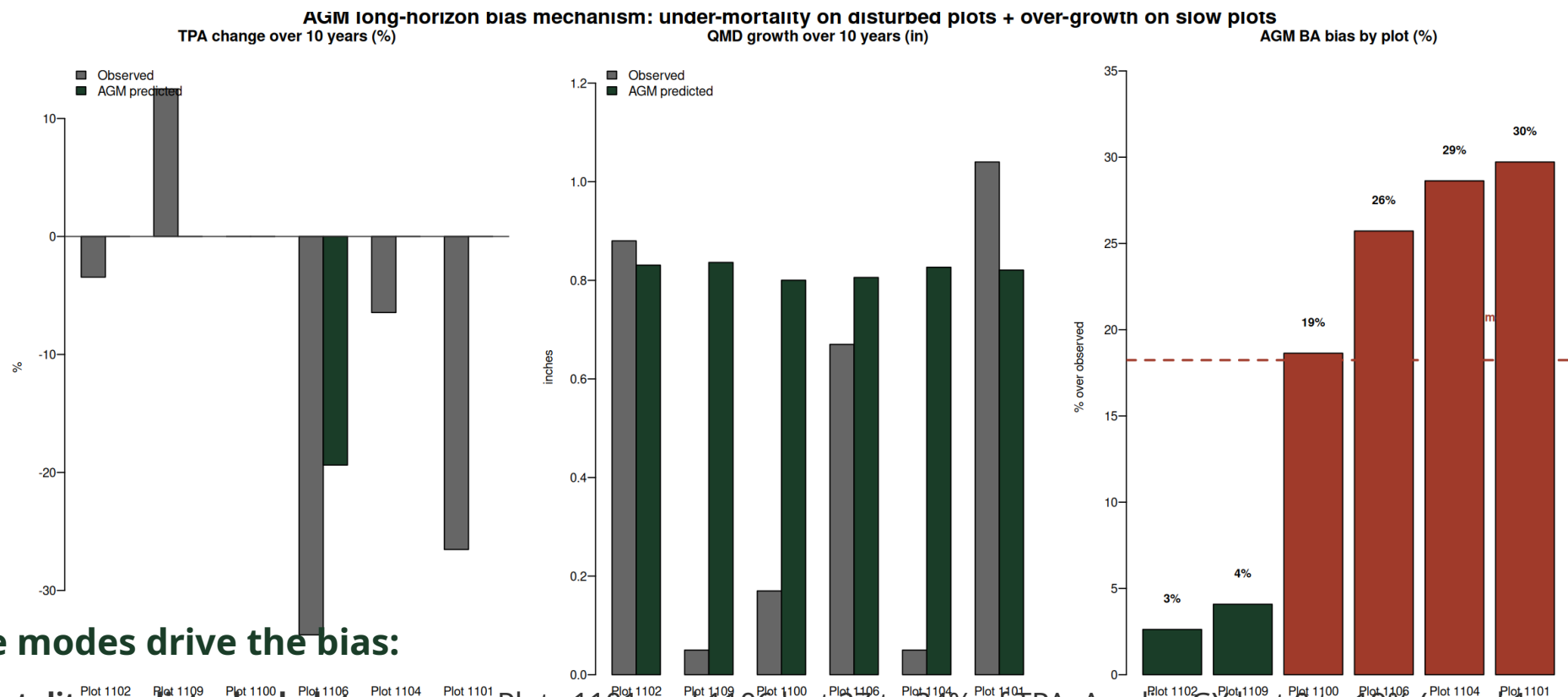
The +5% to +6% AcadianGY and FVS-ACD CFI biases are consistent with the +0% bias the same calibration achieves on 12,029 FIA plots when stand structure differences are accounted for. The CFI biases reflect SILC's Acadian Matrix configuration, not a model failure.

*FIA recompute: SLURM 10591333; calibration/output/comparisons\_overstory/FINDINGS.md*

**Small n CFI scorecard is consistent with the large n FIA validation**

# Where the +18.5% AcadianGY bias actually lives

*Two mechanisms drive the long-horizon over-projection: under-mortality + uniform DBH growth.*



## Two failure modes drive the bias:

- Under mortality on disturbed plots:
- Over growth on slow plots:
- Verified, not a calculation issue:

Plots 1101 and 1106 lost 27 to 34% of TPA; AcadianGY lost 0 to 19% (panel 1, red bars in panel 3).

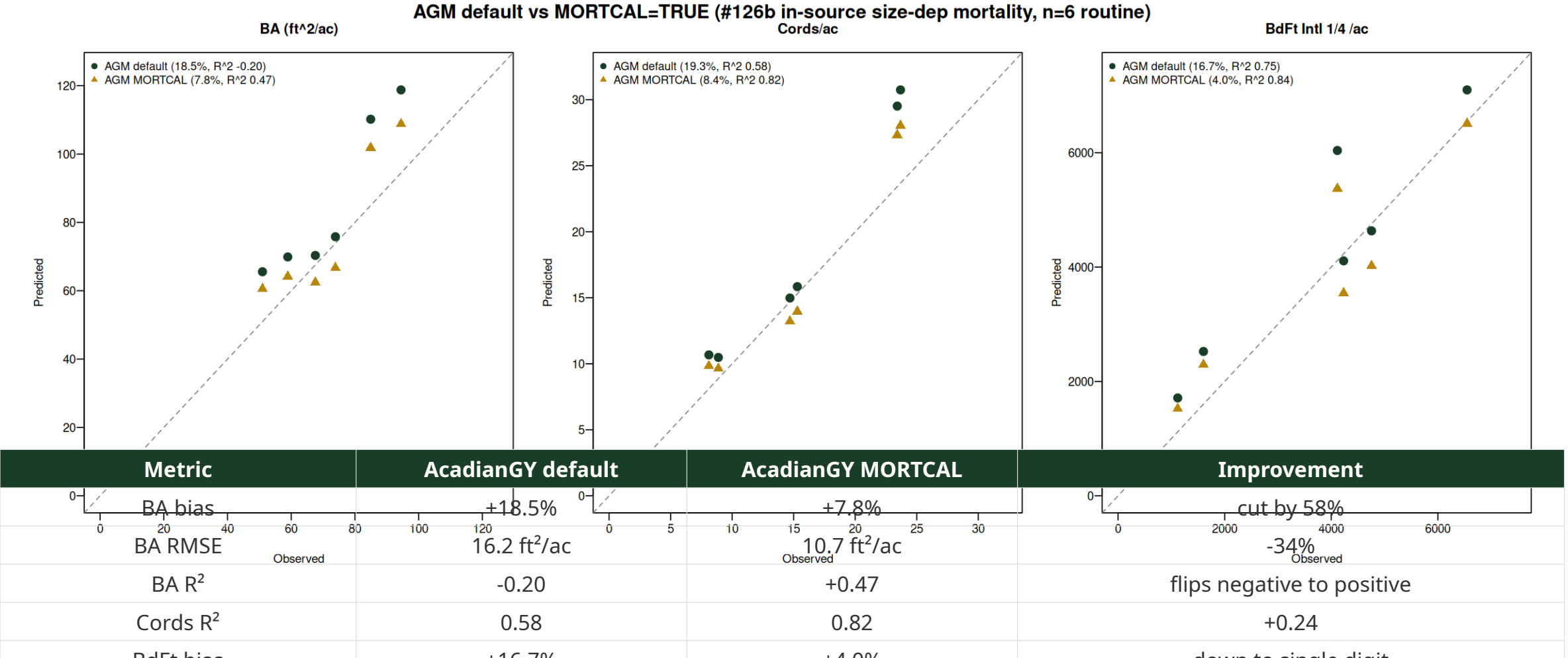
AcadianGY grew QMD a uniform +0.8 in regardless of site; plots 1100 and 1104 grew only 0.05 to 0.17 in observed (panel 2).

BA / Cords / BdFt pipeline reproduces from the AcadianGY treelist to 1e-13 ft<sup>2</sup>/ac. The bias is

**Bias is mechanism, not bug: under mortality + over growth on edge cases**

# Turning on AcadianGY MORTCAL cuts long horizon bias by more than half

#126b size-dependent mortality patch (Maine FIA calibration). Default OFF in 12.3.9; enable via `ops$MORTCAL=TRUE`.



AcadianGY with MORTCAL is the new long horizon champion across BA, cords, sawlog BdFt

# Next steps

1

## **Confirm the 5×2 stratification with SILC**

Sign off that the species composition rules and density threshold match your operational categories

2

## **Backfill the four empty cells with plot level AcadianGY MORTCAL**

Cedar/A+B, Mixedwood/A+B, both Other Softwood; pull from individual SILC CFI plots assigned to those cells, run AcadianGY MORTCAL 100 yr

3

## **Validate year 100 BA against SILC operational expectations**

MORTCAL 100 yr BA settles at 80 to 110 ft<sup>2</sup>/ac. Confirm this matches SILC's expected carrying capacity for managed Acadian stands

4

## **Set the uncertainty bands and ship the operational handoff**

Use AcadianGY MORTCAL RMSE 10.7 ft<sup>2</sup>/ac BA at 10 yr as the per plot uncertainty. Year 100 cords/ac estimates: 36 to 39 across strata

**Four model scaffold is live, awaiting SILC sign off**

# Questions and discussion

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**Aaron R. Weiskittel**

[aaron.weiskittel@maine.edu](mailto:aaron.weiskittel@maine.edu)

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